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Machine Learning for Compensating Velocity Measurement Gaps in Water Flow Time Series Data

Bachelor's thesis

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Author’s declaration of originality

I hereby certify that I am the sole author of this thesis. All the used materials, references to the literature and the work of others have been referred to. This thesis has not been presented for examination anywhere else.

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Abstract

This thesis investigates the use of machine learning for correcting Hydromast water flow velocity measurements toward Acoustic Doppler Velocimeter (ADV) reference measurements. The problem is that raw Hydromast velocity values may differ systematically from ADV measurements, especially in low-velocity conditions. Since Hydromast is suitable for continuous low-cost monitoring, correcting its measurements can improve its practical usefulness.

The aim of the thesis is to develop and evaluate machine learning models that produce corrected Hydromast velocity estimates closer to ADV reference values. The data is cleaned, synchronized, resampled, aligned, and transformed into a supervised learning dataset. Baseline, linear, tree-based, threshold-aware, and Long Short-Term Memory models are evaluated using MAE, RMSE, MBE, and R^2.

The results show that raw Hydromast measurements overestimate ADV velocity on the evaluated test days. Machine learning correction reduces both error and bias. The best overall performance is achieved by tree-based models, while the threshold-aware model gives only a small additional benefit in some low-velocity cases. The LSTM model improves over raw Hydromast but does not outperform the best tree-based models.

The thesis concludes that machine learning can improve Hydromast velocity measurements under the evaluated conditions, but ADV remains necessary for reference measurements, validation, and recalibration.

This thesis is written in English and is 37 pages long, including 6 chapters, 7 figures and 7 tables.

Lühikokkuvõte

Masinõpe Voolukiiruse Mõõtelünga Kompenseerimiseks Veevoolu Aegridade Andmetes

See lõputöö uurib masinõppe kasutamist Hydromasti veevoolu kiiruse mõõtmiste korrigeerimiseks akustilise Doppleri kiirusmõõturi (ADV) võrdlusmõõtmiste suhtes. Probleem seisneb selles, et Hydromasti toored kiirusväärtused võivad ADV mõõtmistest süstemaatiliselt erineda, eriti madala voolukiiruse tingimustes. Kuna Hydromast sobib pidevaks ja madalate kuludega seireks, võib selle mõõtmiste korrigeerimine suurendada seadme praktilist kasutatavust.

Lõputöö eesmärk on välja töötada ja hinnata masinõppemudeleid, mis annavad Hydromasti mõõtmiste põhjal korrigeeritud kiirushinnanguid, mis on lähemal ADV võrdlusväärtustele. Andmed puhastatakse, sünkroniseeritakse, diskretiseeritakse, joondatakse ning teisendatakse juhendatud masinõppe andmestikuks. Hinnatakse baasmeetodeid, lineaarseid mudeleid, puupõhiseid mudeleid, lävendipõhist kaheetapilist mudelit ja pika lühimälu mudelit. Mudelite tulemuslikkust hinnatakse MAE, RMSE, MBE ja R^2 abil.

Tulemused näitavad, et Hydromasti toored mõõtmised hindavad ADV kiirust vaadeldud testipäevadel üle. Masinõppel põhinev korrigeerimine vähendab nii viga kui ka nihet. Parima üldise tulemuslikkuse saavutavad puupõhised mudelid, samas kui lävendipõhine mudel annab mõnel madala kiiruse juhul ainult väikese lisaparanduse. Pika lühimälu mudel parandab tulemusi võrreldes Hydromasti toormõõtmistega, kuid ei ületa parimaid puupõhiseid mudeleid.

Lõputöös järeldatakse, et masinõpe võib parandada Hydromasti veevoolu kiiruse mõõtmisi hinnatud tingimustes, kuid ADV on endiselt vajalik võrdlusmõõtmisteks, valideerimiseks ja edasiseks ümberkalibreerimiseks.

Lõputöö on kirjutatud inglise keeles ning sisaldab teksti 37 leheküljel, 6 peatükki, 7 joonist, 7 tabelit.

List of abbreviations and terms

ADV	Acoustic Doppler Velocimeter
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MBE	Mean Bias Error
R^2	Coefficient of determination
RMSE	Root Mean Square Error
XGBoost	Extreme Gradient Boosting

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1 Introduction

Reliable water flow velocity measurements are important for understanding how water movement changes over time. Such measurements are used in environmental monitoring, coastal and river studies, hydrological analysis, infrastructure planning, and long-term observation of natural water systems. Accurate velocity data helps researchers and organisations evaluate flow conditions, detect changes in the environment, and support decisions related to water management and engineering [1].

However, collecting accurate long-term velocity measurements is challenging because available sensors differ in accuracy, cost, energy consumption, and suitability for continuous deployment. High-accuracy sensors, such as acoustic Doppler velocimeter (ADV), can provide reliable measurements, but they are often expensive and less practical for long-term continuous monitoring [2]. Lower-cost sensors, such as Hydromast, are more suitable for wider and longer deployment, but their raw measurements may require correction before they can be used as reliable velocity estimates [1].

This thesis focuses on correcting Hydromast velocity measurements using ADV measurements as the reference. In this thesis, velocity measurement gaps refer to low-velocity ranges where ADV can still provide reference measurements, but Hydromast raw measurements become unreliable or nearly constant. In this context, the main task is Hydromast-to-ADV velocity measurement correction. The purpose is to evaluate whether machine learning models can transform Hydromast measurements into corrected velocity values that better approximate ADV reference measurements.

The topic also has practical and economic relevance. Continuous water flow monitoring can require long measurement periods and multiple measurement locations, which makes the cost, energy consumption, and maintainability of sensors important. If a lower-cost and lower-energy sensor such as Hydromast can be corrected toward ADV-like reference values, it may become more useful for continuous monitoring tasks. This can support more cost-efficient data collection for environmental monitoring, marine research, coastal engineering, and organisations that need reliable long-term water flow information.

1.1 Problem statement

The main problem addressed in this thesis is that raw Hydromast velocity measurements do not always correspond to ADV reference measurements. ADV provides more reliable velocity measurements, but it is less suitable for continuous long-term monitoring because of its higher cost and energy consumption. Hydromast is more suitable for continuous measurements, but its raw velocity values may contain systematic differences compared with ADV. Therefore, Hydromast measurements need to be corrected before they can be used as ADV-like velocity estimates in the evaluated measurement conditions.

This thesis addresses this problem by developing and evaluating machine learning-based correction methods that transform Hydromast measurements into corrected velocity values closer to ADV reference data.

1.2 Objectives and research questions

The aim of this thesis is to develop and evaluate machine learning models that use Hydromast measurements as input and produce corrected water flow velocity values that more closely approximate ADV measurements.

To achieve this aim, the thesis has the following objectives:

1. to prepare and align Hydromast and ADV time-series data for supervised machine learning;
2. to define an ADV-based target variable for Hydromast-to-ADV velocity measurement correction;
3. to compare raw Hydromast measurements with corrected velocity estimates;
4. to evaluate baseline, linear, tree-based, threshold-aware, and Long Short-Term Memory (LSTM)-based correction models;
5. to assess model performance using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Bias Error (MBE), and the coefficient of determination (R^2);
6. to examine whether correction performance differs in low-velocity conditions.

The thesis addresses the following research questions:

1. Can machine learning models reduce the error and bias of Hydromast velocity measurements compared with the raw Hydromast baseline?
2. Which model type provides the best Hydromast-to-ADV velocity measurement correction performance?
3. How well do the correction models perform in low-velocity conditions?

The scope includes data preparation, sensor alignment, supervised machine learning, evaluation against ADV, and comparison with selected baseline methods. The thesis does not include real-time deployment, hardware redesign, production integration, or full hydrological modelling.

The expected outcome is a reproducible preprocessing and modelling workflow for Hydromast velocity measurement correction. The thesis should show whether the tested correction models reduce error compared with raw Hydromast measurements. The expected outputs include trained models, evaluation metrics, comparison plots, and documented preprocessing steps.

1.3 Thesis structure

Chapter 1 introduces the topic, problem statement, objectives, research questions, scope, and expected outcome. Chapter 2 presents the background, related work, and dataset description. Chapter 3 explains the methodology. Chapter 4 presents the results. Chapter 5 analyses and discusses the results, practical value, reliability, and limitations. Chapter 6 concludes the thesis and outlines future work.

2 Background

This chapter presents the background needed for Hydromast-to-ADV velocity measurement correction. It introduces water flow velocity measurement, the characteristics of ADV and Hydromast sensors, machine learning for sensor correction, preprocessing requirements, and the dataset used in this thesis.

2.1 Water flow velocity measurement

Water flow velocity time-series data describes how water movement changes over time at a measurement location. Depending on the measurement device, velocity may be represented as a single speed value, as horizontal velocity components, or as three-dimensional velocity components. Such measurements are important in hydrology, sediment transport studies, aquatic habitat analysis, flood warning systems, and the validation of numerical models. Egerer et al. state that field measurements of coastal and river velocities are important for both scientific and industrial applications, and that near-bed velocity estimates are especially relevant for sediment transport and river habitat studies [1].

For the purposes of this thesis, the quality of velocity data depends on more than accuracy alone. Sampling frequency, measurement continuity, reliability, and environmental conditions also affect whether sensor data can be used for correction modelling.

2.2 ADV sensor characteristics

Acoustic Doppler velocimeters are commonly used for water velocity measurement in field and laboratory conditions. In this thesis, ADV is treated as the reference sensor because it provides more accurate velocity measurements than Hydromast. This is consistent with the Hydromast calibration study, where ADV was selected as the reference because it is a commercially available standard device for flow measurement [1].

In the thesis dataset, ADV provides the reference measurements but is available only

during burst periods rather than as a fully continuous time series. The exact dataset structure and measurement windows are described in Section 2.7.

The main practical limitations of ADV are its high cost and high energy consumption. Because of these limitations, ADV is often used only during shorter measurement periods rather than as a continuously operating sensor. ADV measurements are also not completely free from uncertainty. Voulgaris and Trowbridge show that ADV can be used effectively for velocity and turbulence measurements, but that uncertainty and noise must still be considered [3]. Therefore, ADV is suitable as the reference target for supervised learning, but the results should still be interpreted with awareness that the reference sensor also has measurement uncertainty.

2.3 Hydromast sensor characteristics

Hydromast is a lower-cost and lower-energy sensor designed for continuous near-bed flow monitoring. The Hall-effect-based version of Hydromast estimates flow velocity and direction from the movement of a flexible mast. The upgraded design allows instantaneous flow estimation and near-real-time data transfer, which makes it suitable for long-term field monitoring [1].

A key advantage of Hydromast is that it records data at 50 Hz. Egerer et al. describe that the device can provide raw data at 50 Hz, consumes approximately 0.15 W, and had a component cost of about 500 euros at the time of writing [1]. These characteristics make Hydromast attractive for continuous and distributed measurements, especially in situations where using many expensive ADV sensors would not be practical. However, Hydromast also has limitations. Its measurements depend on calibration, mast length, flow conditions, and valid measurement range. In the Gulf of Riga study, the validated velocity range of the Hydromast was reported as 0.15–1.0 m/s [2]. The same study explains that at flow speeds below 0.1 m/s, the mast is almost still, the output is almost constant, and the resulting flow-speed estimates are not reliable enough for accurate velocity measurement [2]. Therefore, Hydromast is suitable for continuous monitoring, but its raw values still require validation and possible correction, especially near and below the lower end of the formal validity range.

2.4 Comparison of ADV and Hydromast data

ADV and Hydromast have complementary strengths. ADV provides reliable reference measurements but is less practical for continuous deployment, while Hydromast is more suitable for continuous monitoring but may differ from ADV, especially near measurement limits. Previous field studies show that Hydromast can agree well with ADV after calibration, although limitations remain at low velocities and out-of-range conditions [1]. Therefore, overlapping ADV–Hydromast periods provide the training and evaluation basis for correction models.

2.5 Machine learning for sensor correction and calibration

Machine learning is suitable for this thesis because the relationship between Hydromast and ADV measurements may not be completely linear or constant. Hydromast measurements can be influenced by measurement range, turbulence, noise, calibration settings, and sensor-specific behaviour. A supervised regression model can learn the relationship between Hydromast input features and ADV reference velocity from overlapping measurement periods.

Different regression model families are relevant for this task. Linear and regularized linear models provide simple and interpretable correction functions. Lasso regression is especially useful when feature selection or shrinkage is needed, because it can reduce some coefficients toward zero [4]. Tree-based ensemble models are also relevant because they can model non-linear relationships and interactions between variables. Random forests combine many decision trees and have been widely used for regression and prediction tasks [5]. Gradient boosting builds models sequentially so that later models correct errors made by earlier ones [6]. Extreme Gradient Boosting (XGBoost) is an efficient and scalable implementation of gradient tree boosting that is commonly used for structured tabular data [7]. For sequence-based modelling, long short-term memory networks can be used because they are designed to learn dependencies in sequential data [8].

Research on low-cost sensor calibration supports this approach. Zimmerman et al. used random forests to improve low-cost sensor measurements using reference instruments [9]. Nowack et al. compared several machine learning methods for calibrating low-cost

sensors and showed that both linear and non-linear approaches can be useful depending on the data and transferability requirements [10]. Although these studies focus on air-quality sensors rather than water flow sensors, the structure of the problem is similar: a cheaper sensor is corrected using a more reliable reference sensor. The specific models selected for this thesis are described in the methodology chapter.

2.6 Time-series preprocessing and feature engineering

Preprocessing is necessary before Hydromast and ADV measurements can be compared or used for supervised learning. Because the two sensors use different sampling frequencies and may have timing offsets, the time series must be synchronized, resampled, and aligned so that model inputs and target values refer to comparable time intervals.

The importance of alignment is shown clearly in the Gulf of Riga study [2]. The authors upsampled ADV data to account for different sampling rates and aligned the records by shifting the ADV signal to maximize correlation with Hydromast pressure records. They also excluded records where reliable alignment was not possible [2]. This shows that alignment is not only a technical step, but a requirement for valid comparison between the sensors.

Feature engineering can then be used to represent short-term behaviour of the Hydromast signal and reduce the influence of noise. The detailed feature engineering procedure is described in the methodology chapter, while the dataset structure and data splitting strategy are described in Section 2.7.

2.7 Dataset description

This section describes the experimental data used in the thesis, including the measurement setup, raw datasets, ADV measurement windows, data quality issues, target variable, and train/validation/test split. The focus is on dataset structure and limitations rather than general sensor background.

2.7.1 Measurement setup

The data used in this thesis comes from the same measurement campaign as source [2]. The measurements were collected near Skulte in the eastern Gulf of Riga, Baltic Sea, during August and September 2022. The experiment was conducted in shallow nearshore conditions at approximately 4 m water depth. This environment is relevant because near-bed water velocity can vary over time due to wave-driven motion and local hydrodynamic conditions [2].

The original measurement setup included nine Hydromast devices and one acoustic Doppler velocimeter. The Hydromasts were installed on a 20 × 20 m frame, with 10 m spacing between neighbouring devices. The ADV was placed near the central Hydromast H05. The ADV velocity measurements were taken approximately 0.4 m above the seabed, which corresponds to the centre of the Hydromast vibrating mast [2]. This placement is important for the present thesis because the model is based on comparing the H05 Hydromast signal with the ADV reference signal.

Although the ADV and H05 Hydromast were not identical instruments, their close placement means that they measured comparable near-bed flow conditions. This comparability is necessary for supervised correction. If the two sensors measured substantially different flow fields, differences between their values could not be interpreted mainly as sensor-related differences. In this thesis, the assumption is that the two signals are comparable enough to learn a correction from Hydromast measurements to ADV-like velocity values.

2.7.2 Raw data description

Two main raw datasets are used in this thesis. The ADV dataset uses Unix timestamps and covers the period from `1659452487.375` to `1662625873.875`. Its available columns are `Time [unix]`, `velocity [m/s]`, `Water level [m]`, and `Direction [deg from north]`. During preprocessing, the Unix timestamps are converted into local datetime format using the Europe/Tallinn time zone.

The Hydromast dataset uses datetime timestamps and covers the period from `2022-08-03 00:00:00.000` to `2022-09-07 23:59:59.980`. Its available columns are `time`, `H05`, `depth(m)`, `direction`, `pressure(Pa)raw`, `pressure_norm`, and `temperature`. The

`H05` column is the main Hydromast velocity-related variable used in this thesis. The other variables provide additional information about water depth, flow direction, pressure, normalized pressure, and temperature.

A key practical difference between the datasets is their sampling frequency. ADV measures at 8 Hz, while Hydromast measures at 50 Hz. Therefore, the raw measurements cannot be directly paired row by row. Before modelling, the signals must be converted to a common temporal structure. In the preprocessing workflow, the aligned supervised dataset is created at a 0.5-second resolution.

2.7.3 ADV measurement windows

ADV is not available as a fully continuous signal in the same way as Hydromast. Eelsalu et al. [2] state that the ADV recorded two-minute bursts at 8 Hz every half hour during the measurement period [2], while Hydromast recorded continuously at 50 Hz. Therefore, supervised learning can use only periods where both sensors have available measurements. During these overlapping periods, Hydromast values can be paired with ADV reference values. Outside these periods, Hydromast may still provide continuous measurements, but there is no direct ADV target value for training or evaluation.

2.7.4 Data quality

The raw data contains several quality issues that must be addressed before modelling. Eelsalu et al. [2] describe that raw data were filtered, spurious spikes were removed, and some ADV bursts were excluded because they represented deployment, recovery, or unreliable disturbances [2]. Eelsalu et al. also notes that reliable alignment was not always possible, especially during very low wave-energy conditions.

The data exploration shows that ADV and Hydromast values can differ noticeably in magnitude. In the prototype day used for exploration, Hydromast values stayed close to a lower boundary, while ADV showed lower velocity values. This supports the need for sensor correction in addition to normal preprocessing steps such as timestamp conversion, resampling, alignment, and removal of invalid or unusable observations.

2.7.5 Target variable definition

After preprocessing and temporal alignment, ADV velocity is used as the modelling target. The raw ADV file contains the column velocity [m/s], so the target is treated as velocity magnitude or water speed rather than separate velocity components.

Three ADV-derived target candidates are evaluated in this thesis:

- adv_velocity_last, which represents the latest ADV velocity value within the aligned interval;
- adv_velocity_mean_0p5s, which represents the mean ADV velocity over the 0.5-second aligned interval;
- adv_velocity_mean_1s, which represents a smoother 1-second ADV velocity mean.

The final target column is not selected manually before modelling. Instead, all three target candidates are tested during model development using the same training and validation split. The target that gives the best validation performance, primarily according to MAE and secondarily according to RMSE and bias, is selected as the final target for the main model comparison. This approach is used because the three target definitions represent slightly different ways of summarizing ADV velocity, and the most suitable target should be determined from validation performance rather than assumed in advance.

2.7.6 Data splitting strategy

The processed dataset is divided into training, validation, and test sets using a day-level split. This means that all rows from the same day are kept in only one dataset. The data split uses 36 overlapping days in total: 28 days for training, 4 days for validation, and 4 days for testing. The split is random by day, but a fixed random state of 42 is used so that the split is reproducible.

The resulting datasets contain 290,304 rows for training, 41,472 rows for validation, and 41,472 rows for testing. The validation days are 20220824, 20220814, 20220817, and 20220906. The test days are 20220805, 20220804, 20220816, and 20220811. The remaining 28 days are used for training.

A day-level split is used to reduce data leakage because neighbouring time-series rows are highly related. If individual rows were split randomly, very similar observations from the same measurement period could appear in both training and test sets. By keeping full days separate, the evaluation better tests whether the model can generalize to unseen measurement days. The training set is used to fit models, the validation set is used for model selection and tuning, and the test set is reserved for final evaluation.

2.8 Summary of the research gap

The literature shows that accurate and continuous water flow velocity measurement remains difficult in field conditions. Existing Hydromast studies demonstrate validation against ADV, but limitations remain related to measurement range, calibration, low velocities, and preprocessing. The research gap addressed in this thesis is the application of supervised machine learning to transform Hydromast measurements into corrected velocity estimates that better match ADV reference values.

3 Methodology

This chapter describes the methodological process used to build and evaluate Hydromast velocity measurement correction models. It covers preprocessing steps, feature engineering, baseline methods, selected models, training procedure, and evaluation metrics. Final numerical results are not presented in this chapter.

3.1 Overall workflow

The workflow starts with the ADV and Hydromast datasets described in Section 2.7. First, timestamps and variables are standardized, and only days with real Hydromast–ADV overlap are kept. The data is then transformed to a common 0.5-second resolution, aligned by timestamp, and converted into a supervised dataset with Hydromast-based features and ADV-based target values. After feature engineering, the prepared data is split by day into training, validation, and test sets as described in Section 2.7.6.

3.2 Data cleaning

Data cleaning is required because supervised learning depends on reliable input-target pairs. If timestamps, missing values, duplicates, or non-numeric values are not handled correctly, the model may learn from data errors rather than from the actual relationship between Hydromast and ADV measurements.

The preprocessing standardizes column names, converts timestamps to the Europe/Tallinn time zone, and parses required variables as numerical values. Duplicate timestamps are removed, the data is sorted by time, and rows with missing time values or missing required velocity values are dropped. After preprocessing, the final aligned dataset contains no missing values.

3.3 Synchronization and resampling

Because ADV and Hydromast have different sampling rates, the raw rows cannot be directly paired. Both datasets are converted to a common 0.5-second resolution so that Hydromast inputs and ADV target values describe the same time interval.

ADV velocity is aggregated into variables such as `adv_velocity_last` and `adv_velocity_mean_0p5s`. An additional 1-second ADV rolling mean is also created. ADV direction is aggregated using a circular mean and transformed into sine and cosine components to avoid the discontinuity between 0° and 360°. Hydromast variables are aggregated to the same 0.5-second structure, including velocity, depth, pressure, normalized pressure, temperature, and direction. Direction is also transformed into sine and cosine components. After resampling, ADV and Hydromast data are merged by timestamp. ADV segment identifiers are created to prevent rolling features from being calculated across gaps between ADV bursts.

3.4 Feature engineering

Feature engineering is used to describe the Hydromast signal in a way that helps predict ADV-like velocity. The main feature group contains direct Hydromast measurements, including current Hydromast velocity, depth, pressure, normalized pressure, temperature, and direction represented by sine and cosine values. Current Hydromast velocity is the most direct input because the goal is to correct this measurement. Pressure and depth may help describe wave-related conditions, while temperature may be relevant if sensor behaviour changes with environmental conditions.

The second feature group contains rolling statistics of Hydromast velocity. It contains rolling means over several time windows, as well as rolling standard deviation, minimum, and maximum values. These features describe short-term flow behaviour and may reduce the effect of noise. Quantile features over 5-second and 10-second windows are also created to represent the recent distribution of Hydromast values.

The third group contains change and slope features. These describe whether the Hydromast signal is increasing, decreasing, or stable. This is useful because two measurements with the same current value may belong to different flow situations depending on recent changes.

The fourth group contains threshold-based features, such as the fraction of recent Hydromast values below selected velocity thresholds. These features are relevant because Hydromast has known limitations near the lower end of its measurement range [1]. The physical measurement-range threshold reported in the literature was not used as a fixed

exclusion rule in the main dataset, because the aim of this thesis is to evaluate whether machine learning can improve Hydromast velocity measurements also in low-velocity conditions. Instead, low-velocity behaviour is handled through threshold-based features and a separate threshold-aware model. This makes it possible to test whether the models can learn useful correction patterns in the low-velocity region rather than removing those observations before modelling.

3.5 Baseline methods

Baseline methods are required to show whether correction models improve over simpler alternatives. The first baseline is raw Hydromast velocity compared directly with aligned ADV velocity. This baseline represents the original measurement quality before correction. The second baseline is the training-median baseline, where the median ADV target value from the training set is used as a constant prediction for the validation and test sets.

3.6 Machine learning models for Hydromast velocity measurement correction

The selected models are organized from simple and interpretable methods to more flexible non-linear methods. This structure makes it possible to evaluate whether complex models provide a meaningful improvement over simpler correction approaches.

The first model group consists of linear regression models. Ordinary linear regression is used as a simple interpretable correction model. Ridge regression, Lasso regression, and Elastic Net regression are also tested because the engineered feature set contains several rolling and related variables that may be correlated. Lasso regression is included because it performs coefficient shrinkage and can reduce the influence of less useful features [4]. Regularized linear models therefore provide a useful comparison against more flexible non-linear approaches.

The second model group consists of tree-based ensemble models. Random Forest Regressor is included because random forests combine multiple decision trees and can capture non-linear relationships between input features and target values [5]. Random forests are also relevant because similar approaches have been used successfully for low-

cost sensor calibration against reference instruments [9]. Histogram-based Gradient Boosting Regressor is included because gradient boosting builds models sequentially, where later models correct errors made by earlier models [6]. This makes it suitable for tabular regression problems where the Hydromast–ADV relationship may be non-linear. XGBoost Regressor is also tested as an additional gradient boosting implementation because it is designed as a scalable and efficient tree boosting system [7]. Since Nowack et al. show that both linear and non-linear algorithms can be relevant in low-cost sensor calibration, comparing these model groups is appropriate for the present correction task [10].

The third model type is a threshold-aware two-stage model. This model is included because Hydromast has known limitations near the lower end of its measurement range [1]. In this approach, the data is divided into lower-velocity and normal-velocity regions using a threshold selected from the training data. Separate regression models are then trained for these regions. The purpose is to test whether low-velocity correction improves when the model explicitly treats this range as a special case.

The fourth model group consists of neural network models. An LSTM model is included because the data is time-series data and recent Hydromast measurements may contain useful temporal information for predicting ADV-like velocity. LSTM networks are designed to learn dependencies in sequential data and can retain information across longer input sequences than basic recurrent neural networks [8]. Unlike tabular models that use manually engineered rolling features, the LSTM model receives short sequences of Hydromast-based observations and learns temporal dependencies directly. Since the experiments are run without GPU acceleration, the LSTM search space is kept limited to avoid excessive computation time.

The final model comparison includes only models that are successfully trained and evaluated on the same validation and test split. This ensures that all reported results are comparable.

3.7 Model training and hyperparameter tuning

The experimental workflow is implemented in Python using Visual Studio Code. The experiments are run on an Asus Zenbook UX3404VA laptop with a 13th Gen Intel(R)

Core(TM) i9-13900H processor, 32 GB RAM, a 64-bit system architecture, and Windows 11 Home 25H2. The graphics card is not used for model training. The software environment includes Visual Studio Code version 1.119.0 and Python version 3.13.12. The main Python libraries used for data processing, modelling, and visualization were pandas, numpy, matplotlib, scikit-learn, xgboost, and tensorflow. Additional standard Python modules, including pathlib, csv, warnings, sys, time, itertools, and zoneinfo, were used for file handling, runtime control, iteration, and time zone processing.

Model training uses the day-level train, validation, and test split described in Section 2.7.6. The training set is used to fit model parameters, the validation set is used for target selection, model comparison, and hyperparameter tuning, and the test set is kept separate for final evaluation. The test set is not used during model selection. This separation is necessary because the test set should represent unseen measurement days.

The first modelling step is target selection. For each of the three candidate target columns, the same baseline models are trained and evaluated on the validation set. The target column with the best validation MAE is selected for the main experiment. RMSE and MBE are also checked to ensure that the selected target does not produce unusually large errors or systematic bias.

After the target is selected, hyperparameter tuning is performed on the validation set. Since the data is time-series data split by day, random row-level cross-validation is not used. Instead, the predefined validation set is used to avoid leakage between neighbouring time points. The same random seed is used where supported by the model so that the experiments are reproducible.

The hyperparameter search spaces are chosen according to the characteristics of the selected model families. Regularized linear models are tuned mainly through the regularization strength, while random forest and gradient boosting models are tuned through parameters controlling tree depth, number of estimators, learning rate, sampling, and regularization [4], [5], [6], [7]. The LSTM model is tuned through sequence length, hidden units, number of layers, dropout, learning rate, and batch size, because these parameters affect how temporal dependencies are learned from sequential input data [8].

For linear models, the following settings are tested:

Table 3.1. Hyperparameter search space for linear models.

Model	**Hyperparameters tested**
Linear Regression	No main hyperparameters
Ridge Regression	alpha: 0.01, 0.1, 1, 10, 100
Lasso Regression	alpha: 0.0001, 0.001, 0.01, 0.1, 1
Elastic Net	alpha: 0.0001, 0.001, 0.01, 0.1, 1 l1_ratio: 0.2, 0.5, 0.8

For tree-based ensemble models, the following search spaces are used:

Table 3.2. Hyperparameter search space for tree-based ensemble models.

Model	**Hyperparameters tested**
Random Forest Regressor	n_estimators: 200, 500 max_depth: None, 10, 20 min_samples_leaf: 1, 5, 10 max_features: sqrt, 0.5, 1.0
HistGradientBoostingRegressor	learning_rate: 0.03, 0.05, 0.1 max_iter: 200, 500 max_leaf_nodes: 15, 31, 63 l2_regularization: 0, 0.01, 0.1 min_samples_leaf: 20, 50
XGBoost Regressor	n_estimators: 300, 500, 800 learning_rate: 0.03, 0.05, 0.1 max_depth: 3, 5, 7 subsample: 0.7, 0.9, 1.0 colsample_bytree: 0.7, 0.9, 1.0 reg_lambda: 1, 5, 10

For the threshold-aware two-stage model, threshold candidates are selected from the training data using empirical Hydromast velocity quantiles. The tested threshold candidates are the 5th, 10th, 15th, and 20th percentiles of training-set Hydromast velocity. For each threshold, separate models are trained for the low-velocity and normal-velocity subsets. The best threshold is selected according to validation MAE, with additional attention to low-velocity performance.

For the LSTM model, input sequences are created only after the day-level split has been performed, so that sequences do not cross from one dataset into another. The following settings are tested:

Table 3.3. Hyperparameter search space for LSTM model.

Model	Hyperparameters tested
LSTM	sequence length: 10, 20, 60 time steps hidden units: 32, 64 number of LSTM layers: 1, 2 dropout: 0, 0.2 learning rate: 0.001, 0.0005 batch size: 64, 128

Because the aligned dataset has a 0.5-second resolution, sequence lengths of 10, 20, and 60 time steps correspond to approximately 5, 10, and 30 seconds of recent Hydromast information. The LSTM is trained with early stopping based on validation MAE. Training is stopped if validation performance does not improve for five consecutive epochs, and the best validation checkpoint is used for evaluation.

Linear and neural network models use standardized numerical features fitted only on the training set. The same transformation is then applied to validation and test data. Tree-based models are trained on the original numerical feature values because scaling is not required for these models.

Within each model group, the best configuration is selected using validation MAE, with RMSE and MBE used as secondary checks. The final comparison includes the raw Hydromast baseline, the training-median baseline, the best linear model, the best tree-based model, the best LSTM model, and the best threshold-aware model. The test set is used only for final reporting and is not used for target selection, hyperparameter tuning, or model selection.

3.8 Evaluation metrics

Predictions are evaluated against the final ADV target selected using the procedure described in Sections 2.7.5 and 3.7. Several metrics are used because each metric describes model performance from a different perspective.

MAE is used as the primary metric. It measures the average absolute error between prediction and reference velocity and is expressed in the same unit as velocity, m/s. This makes it easy to interpret. Willmott and Matsuura argue that MAE is a clear measure of

average model error [11]. In this thesis, MAE directly shows how far corrected Hydromast values are from ADV on average.

RMSE is used as a secondary metric. It also measures error in m/s, but gives stronger weight to larger errors. This is useful because large deviations from ADV may be especially problematic in sensor correction. Chai and Draxler argue that RMSE provides useful information when larger errors matter [12].

Mean Bias Error is used to identify systematic overestimation or underestimation. A model may have low absolute error but still consistently predict values that are too high or too low. MBE helps detect this direction of error.

R^2 is used as a supplementary metric. It shows how much variation in ADV velocity is explained by the model. However, R^2 should not be used alone, because a model may explain variation while still producing unacceptable absolute errors.

If low-velocity evaluation is included, it should be treated as additional analysis because Hydromast limitations at low velocities were identified in the background literature. The main evaluation should still focus on correction performance across the available velocity range.

3.9 Use of generative artificial intelligence

Generative artificial intelligence was used as a supporting tool during the thesis process. ChatGPT versions 5.4 and 5.5 were used to ask programming-related questions, receive suggestions on code structure, check code fragments, understand error messages, and generate non-central utility code, such as plotting code [13]. The tool was also used for general advice during implementation decisions.

The author reviewed, modified, and tested all code before using it in the thesis workflow. Generative AI was not used as a source of experimental data, was not used to fabricate results, and did not replace the original scientific sources cited in the thesis. All factual claims, methodological decisions, and literature-based arguments are based on the cited sources, datasets, and the author's own implementation and analysis.

4 Results

This chapter presents the experimental results of the Hydromast-to-ADV velocity measurement correction workflow. The aim is to evaluate whether correction models can transform Hydromast measurements into velocity estimates that better approximate ADV reference measurements. The experiments were evaluated using the selected target variable, adv_velocity_mean_1s. The chapter focuses on the raw sensor comparison, baseline results, machine learning correction results, low-velocity performance, visual comparisons, and model interpretation.

4.1 Exploratory analysis of raw sensor data

The raw data analysis showed clear differences between ADV and Hydromast measurements before correction. During the explored prototype day, 2022-08-20, 6144 paired one-second observations were available. ADV velocity had a mean of 0.029 m/s, while the Hydromast mean was 0.086 m/s. The average Hydromast-minus-ADV difference was 0.057 m/s, and the RMSE between the raw signals was 0.059 m/s. The correlation between the two signals was 0.297, indicating weak direct linear agreement.

Figure 1 shows a representative raw time-series window before correction. Hydromast velocity was concentrated near a narrow range around approximately 0.085–0.09 m/s, while ADV varied more strongly and often reached lower velocity values. This confirms that the raw difference between the sensors is systematic and cannot be explained only by random noise.

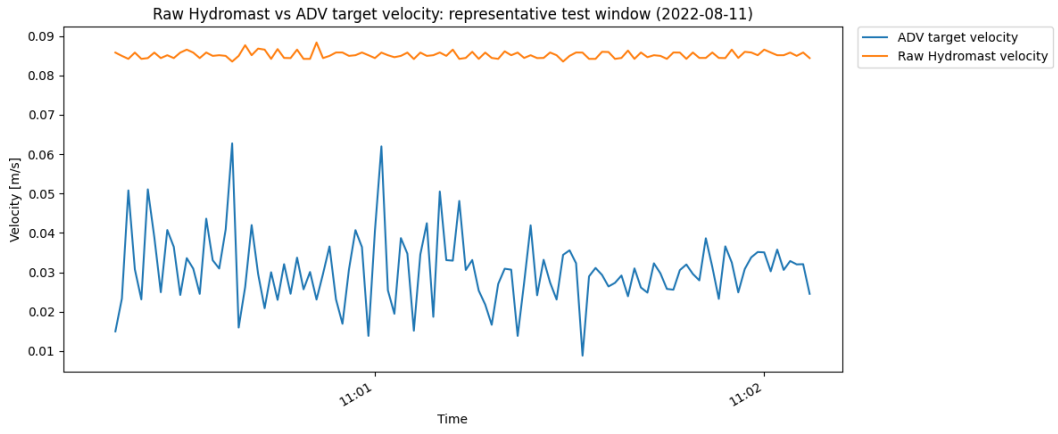


Figure 1. Raw Hydromast and ADV target velocity in a representative test-set window.

The target distribution also differed between data splits. In the training set, adv_velocity_mean_1s had a mean of 0.072 m/s and a standard deviation of 0.076 m/s. In the validation set, the same target had a lower mean of 0.036 m/s and a standard deviation of 0.024 m/s. This distribution difference is relevant for interpreting the validation results, because the validation period contains lower-velocity conditions than the training set.

After common row-level filtering, the row-level comparison used 145,152 training rows, 20,736 validation rows, and 20,736 test rows. The LSTM model used sequence-aligned rows, so its evaluation used fewer rows than the row-level models.

4.2 Raw Hydromast versus ADV comparison

The raw Hydromast baseline establishes the measurement problem before correction. On the test set, the ADV target mean was 0.039 m/s, while the raw Hydromast prediction mean was 0.087 m/s. This produced a positive mean bias error of 0.049 m/s, showing that raw Hydromast strongly overestimated the ADV reference on the test days.

Table 4.1. Raw Hydromast baseline performance.

Split	**MAE [m/s]**	**RMSE [m/s]**	**MBE [m/s]**	**R^2**	**Rows**
Validation	0.056	0.065	0.055	-6.360	20,736
Test	0.050	0.055	0.049	-3.362	20,736

The raw baseline shows that uncorrected Hydromast values are not sufficient as ADV-like velocity estimates in the evaluated data. A useful correction model should therefore reduce MAE, RMSE, and the strong positive bias.

The same pattern is visible in the scatter plot in Figure 2, where raw Hydromast values do not follow the diagonal reference line.

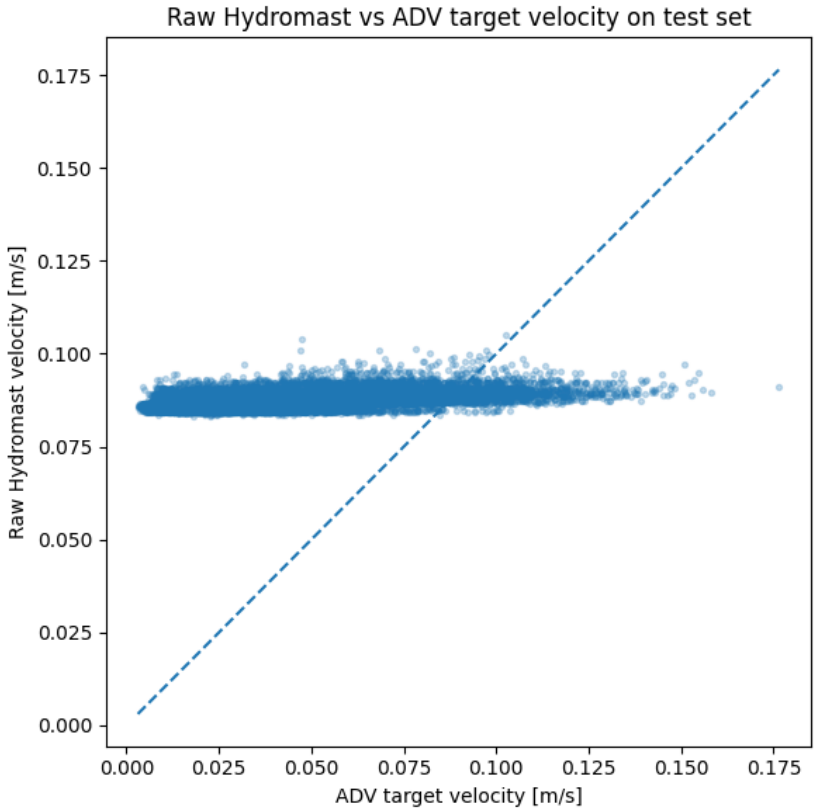


Figure 2. Raw Hydromast velocity compared with ADV target velocity on the test set.

4.3 Baseline model results

Several baseline methods were used to judge whether more advanced correction models provide real improvement. The first baseline is raw Hydromast velocity. The second is a median baseline, where the prediction is the median ADV target value from the training data.

On the test set, the median baseline achieved MAE of 0.024 m/s, which was much lower than the raw Hydromast MAE of 0.050 m/s. This result shows that the raw Hydromast signal was strongly biased upward on the test days. However, the median baseline does not use the Hydromast signal dynamically.

The best linear model improved further, with a test MAE of 0.019 m/s and RMSE of 0.023 m/s. This shows that a simple linear correction using Hydromast-based features already provides useful improvement over both raw Hydromast and the median baseline.

4.4 Machine learning correction results

The final comparison included raw Hydromast, the median baseline, the best linear model, the best tree-based model, the best LSTM model, and the best threshold-aware model. The main comparison is presented in Table 4.2.

Table 4.2. Final validation and test results.

Split	Model	MAE [m/s]	RMSE [m/s]	MBE [m/s]	R^2	Rows
Validation	Raw Hydromast baseline	0.056	0.065	0.055	-6.360	20,736
Validation	Median baseline	0.022	0.026	0.011	-0.204	20,736
Validation	Best linear model	0.021	0.032	0.011	-0.729	20,736
Validation	Best tree-based model	0.018	0.025	0.008	-0.081	20,736
Validation	Best LSTM model	0.020	0.030	0.010	-0.576	17,088
Validation	**Best threshold-aware model**	**0.018**	**0.024**	**0.006**	**-0.033**	**20,736**
Test	Raw Hydromast baseline	0.050	0.055	0.049	-3.362	20,736
Test	Median baseline	0.024	0.028	0.008	-0.103	20,736
Test	Best linear model	0.019	0.023	0.006	0.230	20,736
Test	**Best tree-based model**	**0.017**	**0.022**	**-0.001**	**0.298**	**20,736**
Test	Best LSTM model	0.019	0.023	0.004	0.216	17,088
Test	Best threshold-aware model	0.017	0.022	-0.001	0.290	20,736

Note. The best validation and test MAE results are shown in bold.

The model labels in Table 4.2 refer to validation-selected configurations. The best linear model is Elastic Net Regression. The best tree-based model is a single-stage

HistGradientBoostingRegressor. The best LSTM model is the validation-selected sequence model with a 20-step input sequence. The best threshold-aware model is a two-stage HistGradientBoostingRegressor with a training-set Hydromast low-velocity threshold of 0.085 m/s.

Figure 3 visualizes the MAE values from the final comparison table. This makes the difference between the raw baseline, simple baselines, and correction models easier to compare.

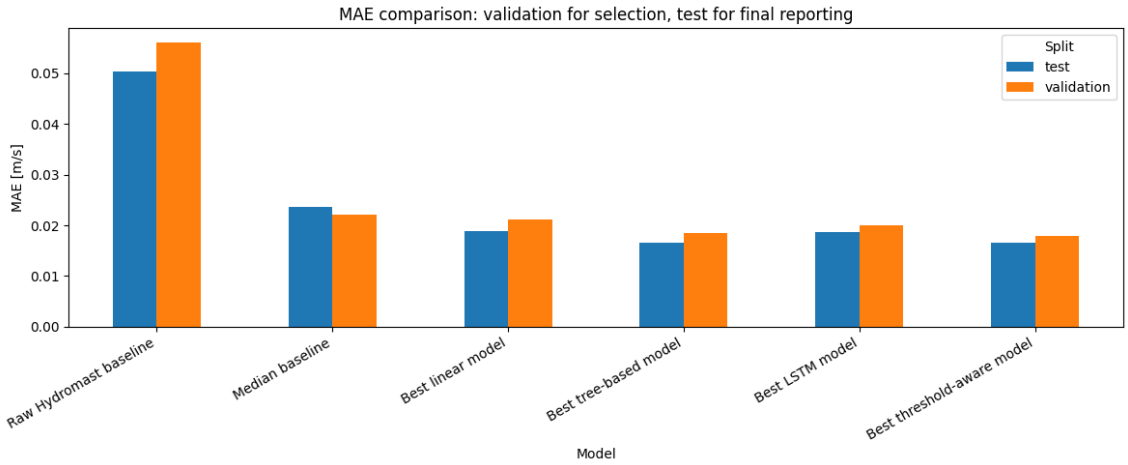


Figure 3. MAE comparison of baseline and correction models on the validation and test sets.

To further examine systematic error, Figure 4 presents the residual distributions on the test set. Residuals are calculated as prediction minus ADV target velocity.

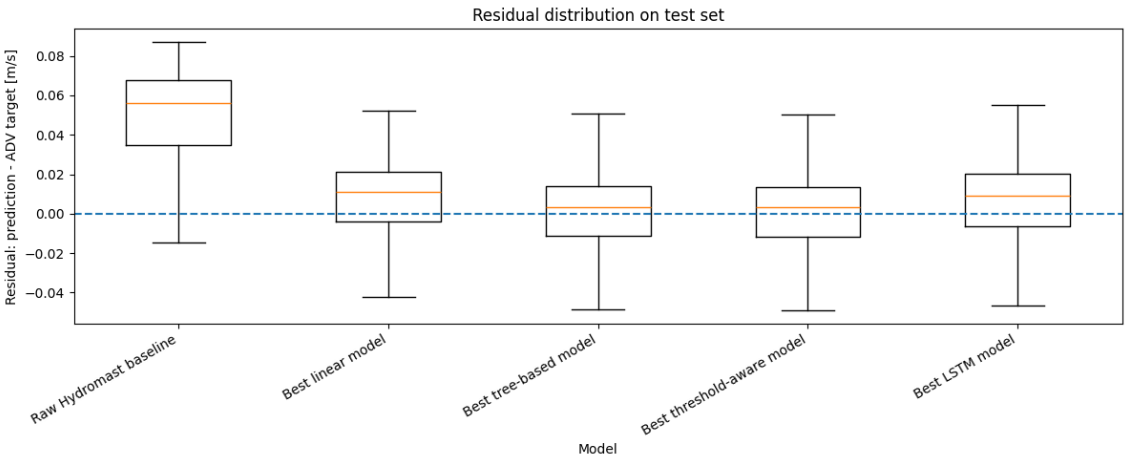


Figure 4. Test-set residual distributions for raw Hydromast and correction models.

The threshold-aware model had the lowest validation MAE, 1.79×10^{-2} m/s. The single-stage tree-based model was very close, with validation MAE of 1.85×10^{-2} m/s. On the test set, the single-stage tree-based model had the lowest MAE, 1.660×10^{-2} m/s, while the threshold-aware model had MAE of 1.663×10^{-2} m/s. The difference was only 3×10^{-5} m/s, so the threshold-aware model should not be described as clearly superior. Instead, the results show that both tree-based approaches strongly improved over raw Hydromast and performed very similarly.

The LSTM model also improved substantially over raw Hydromast, but it did not outperform the tree-based models. On the same LSTM-aligned test subset, the tree-based model achieved MAE of 0.017 m/s, while the LSTM achieved 0.019 m/s. This indicates that, for the available data and implementation, the tree-based model performed better than the sequence model.

4.5 Performance at low velocities

Low-velocity performance is important because previous Hydromast work identifies limitations near the lower end of the measurement range [1]. The experiments therefore included additional analysis for low Hydromast velocity and target-based velocity bins.

Using the threshold selected by the threshold-aware model, the test set was divided into low Hydromast velocity and normal/high Hydromast velocity regions. Both the single-stage tree-based model and the threshold-aware model strongly reduced error in the low-velocity region.

Table 4.3. Test performance in low and normal/high Hydromast velocity regions.

Region	**Model**	**MAE [m/s]**	**RMSE [m/s]**	**MBE [m/s]**	**Rows**
Full test set	Raw Hydromast baseline	0.050	0.055	0.049	20,736
Full test set	Best tree-based model	0.017	0.022	-0.001	20,736
Full test set	Best threshold-aware model	0.017	0.022	-0.001	20,736

Region	Model	MAE [m/s]	RMSE [m/s]	MBE [m/s]	Rows
Low Hydromast velocity	Raw Hydromast baseline	0.058	0.059	0.057	3,834
Low Hydromast velocity	Best tree-based model	0.014	0.017	0.006	3,834
Low Hydromast velocity	Best threshold-aware model	0.013	0.017	0.006	3,834
Normal/high Hydromast velocity	Raw Hydromast baseline	0.049	0.054	0.047	16,902
Normal/high Hydromast velocity	Best tree-based model	0.017	0.023	-0.002	16,902
Normal/high Hydromast velocity	Best threshold-aware model	0.017	0.023	-0.003	16,902

Figure 5 summarizes the full test-set, low Hydromast velocity, and normal/high Hydromast velocity MAE values for the raw baseline and the two strongest correction models.

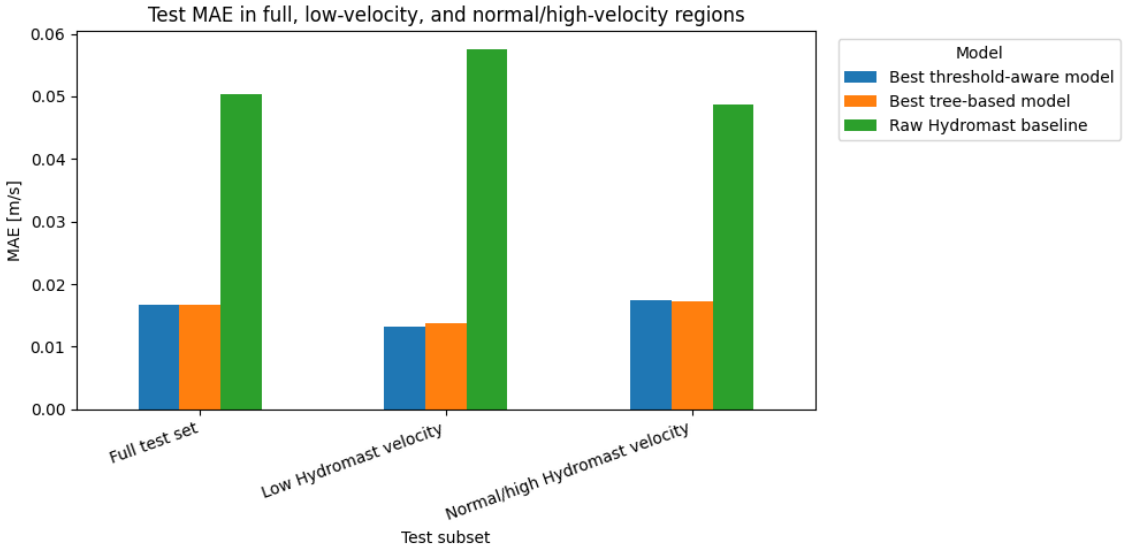


Figure 5. Test MAE in the full test set and Hydromast velocity regions.

The threshold-aware model had the best low Hydromast velocity MAE, 1.31×10^{-2} m/s, compared with 1.37×10^{-2} m/s for the single-stage tree-based model. However, this

improvement was small, and the threshold-aware model was slightly worse than the single-stage tree-based model on the full test set.

A second velocity-bin analysis used ADV target thresholds from the training distribution. On the test set, the threshold-aware model performed best in the low ADV velocity bin, with MAE of 1.34×10^{-2} m/s, while the single-stage tree-based model had MAE of 1.38×10^{-2} m/s. In the medium bin, the LSTM and linear model performed best, with MAE values around 8.8×10^{-3} m/s. In the high bin, raw Hydromast had the lowest MAE, while the correction models tended to underpredict high velocities.

4.6 Visual comparison of corrected time series

The final comparison includes time-series plots comparing ADV target velocity, raw Hydromast velocity, the best tree-based correction, and the threshold-aware correction. In the representative test window, raw Hydromast remained close to a nearly constant high level, while the ADV target varied at lower velocities. The corrected predictions were much closer to the ADV level than the raw Hydromast signal.

Figure 6 compares ADV target velocity, raw Hydromast velocity, and the two strongest corrected predictions in a representative test-set window.

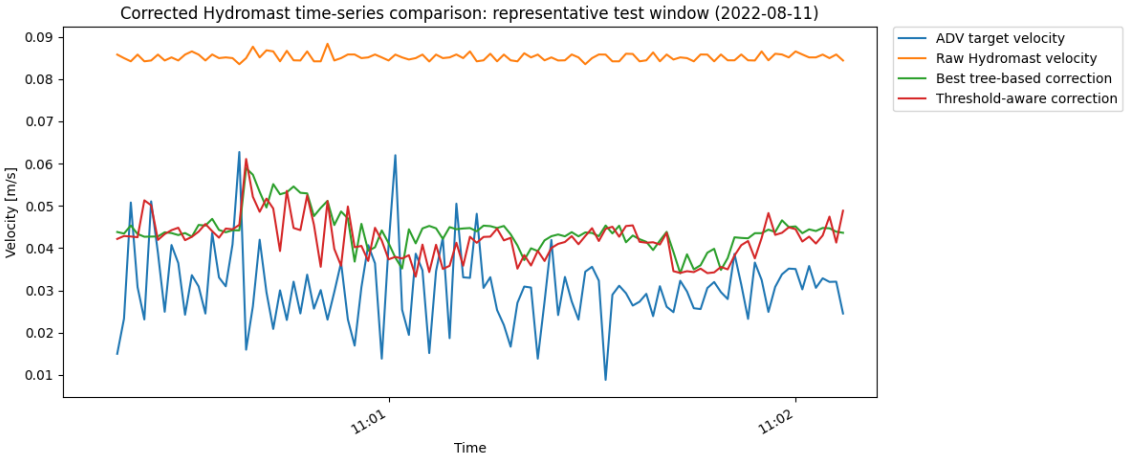


Figure 6. Corrected Hydromast time-series comparison in a representative test-set window.

The corrected time-series plot also shows that the tree-based and threshold-aware models produced very similar predictions. Both models reduced the large positive offset of raw

Hydromast. However, the corrected signals were smoother than the ADV target and did not fully reproduce all short-term ADV fluctuations.

Figure 7 shows the relationship between corrected predictions and ADV target velocity across the test set. The diagonal line represents ideal agreement.

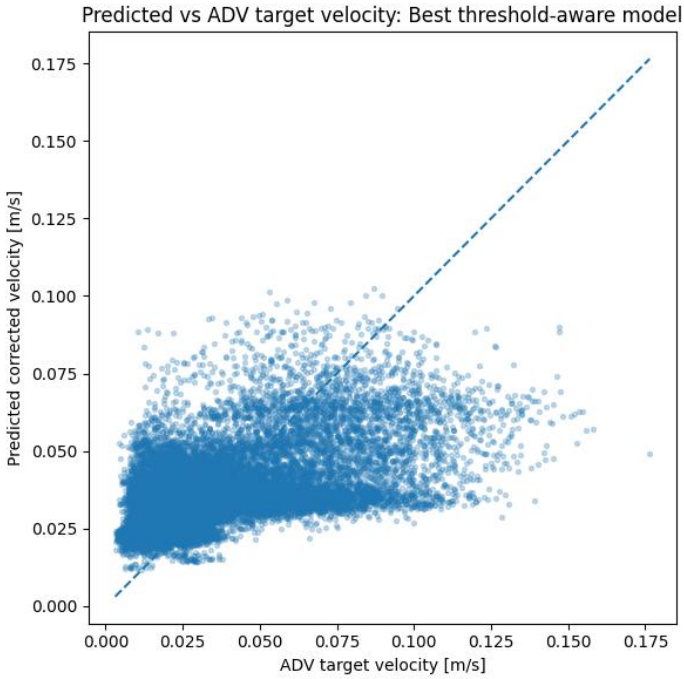


Figure 7. Predicted corrected velocity compared with ADV target velocity for the validation-selected model.

The predicted-versus-ADV scatter plot for the validation-selected threshold-aware model showed a broader and more ADV-dependent prediction pattern than the raw Hydromast scatter plot. Nevertheless, the corrected scatter still showed underprediction in higher-velocity regions, which agrees with the velocity-bin analysis.

4.7 Model interpretation

Model interpretation was performed using permutation feature importance for the best tree-based model. The most important features were identified by measuring how much validation MAE increased when each feature was permuted.

Table 4.4. Top permutation feature importance results.

Rank	Feature	Mean validation MAE increase [10^{-3} m/s]
1	hydromast_depth_m	4.96
2	hydromast_temperature	3.91
3	hydro_mean_20s	2.03
4	hydromast_pressure_raw_pa	1.89
5	pressure_norm_std_10s	1.20
6	hydro_dir_sin_mean_10s	0.84
7	hydro_max_10s	0.57
8	hydro_min_10s	0.19
9	hydro_q90_10s	0.16
10	pressure_norm_std_5s	0.15

The feature importance results suggest that the correction model did not rely only on the current raw Hydromast velocity. Depth, temperature, longer-window Hydromast velocity averages, raw pressure, pressure variability, and direction-related features contributed to the correction. However, permutation importance should be interpreted carefully because it shows predictive usefulness in this dataset, not direct physical causality.

5 Analysis and Discussion

5.1 Interpretation of main results

The results in Chapter 4 show that raw Hydromast velocity measurements differed substantially from the ADV reference measurements on the validation and test days. The raw Hydromast baseline had a strong positive bias, meaning that uncorrected Hydromast values generally overestimated ADV velocity. This confirms the central problem of the thesis: although Hydromast is useful as a continuous and low-cost measurement device, its raw values cannot be treated as directly equivalent to ADV measurements under the evaluated conditions.

The main finding is that supervised machine learning correction reduced the difference between Hydromast and ADV. All correction models improved over the raw Hydromast baseline, showing that Hydromast measurements contained useful information for estimating ADV-like velocity values. This supports the thesis aim, because the experiments demonstrate that machine learning can transform Hydromast-based measurements into corrected estimates closer to the ADV reference.

The best overall results came from tabular tree-based approaches. The threshold-aware model achieved the best validation MAE, while the single-stage tree-based model achieved slightly better test MAE. Since the difference between them was very small, the threshold-aware model should not be presented as clearly superior. A more careful interpretation is that both tree-based approaches performed very similarly, while the threshold-aware structure gave only a small additional benefit in some low-velocity cases.

The LSTM model also improved over raw Hydromast, but it did not outperform the tree-based models. This suggests that, for the available dataset and feature set, engineered rolling and change-based features already represented much of the useful temporal information. The LSTM may require more data, more diverse measurement conditions, or a different architecture to show a stronger advantage.

Overall, the experiments show that machine learning can reduce the velocity difference between Hydromast and ADV. The most reliable results came from tabular tree-based models rather than from the LSTM sequence model.

The feature importance results in Section 4.7 show that the best tree-based model did not rely only on the current Hydromast velocity value. Depth, temperature, the 20-second Hydromast velocity mean, raw pressure, and pressure variability were among the most important features. These variables likely helped the model account for the measurement context around the raw Hydromast value. For example, pressure-related variables may describe wave-related conditions, while rolling Hydromast statistics describe recent flow behaviour. This additional contextual information may be especially relevant in low-velocity conditions, where the instantaneous Hydromast velocity value alone may be less reliable. However, the feature importance results should be interpreted as predictive importance within this dataset, not as proof of direct physical causality.

5.2 Comparison with alternative approaches

The raw Hydromast baseline represented the practical situation before correction. Its poor agreement with ADV showed that the problem could not be solved by using Hydromast values directly as ADV-like estimates. It also made the improvement achieved by correction models easier to interpret.

The median baseline was relatively strong despite its simplicity. This can be explained by the low ADV velocity values in the validation and test sets. Predicting a central training-set value already reduced much of the positive bias of raw Hydromast. However, the median baseline is not a dynamic correction method. It does not respond to changes in Hydromast measurements, pressure, depth, direction, or short-term flow behaviour. Therefore, it is useful as a conservative reference point, but it cannot replace a model that uses measurement features.

Linear models were useful because they showed that part of the Hydromast-to-ADV velocity measurement correction can be captured by a simple relationship between engineered features and the ADV target. However, the stronger performance of tree-based models suggests that this relationship was not fully linear.

Tree-based ensemble methods were well suited to the task because Hydromast behaviour may depend on interacting factors such as velocity range, pressure-related conditions, depth, direction, and short-term variation. These models can capture nonlinear relationships and feature interactions without requiring them to be specified manually [4],

[5], [6]. This likely explains why the best tree-based model achieved the most reliable overall performance.

The LSTM model was included because the data has a time-series structure and recent observations may help predict ADV-like velocity. LSTM networks are designed to learn dependencies in sequential data [8]. However, in this thesis, the LSTM did not outperform the best tabular tree-based models. A likely reason is that engineered rolling features already summarized much of the useful short-term temporal information. Another possible reason is that the available validation and test data were limited, while neural network models often require larger and more diverse datasets.

The threshold-aware model was motivated by known Hydromast limitations near the lower end of its measurement range [1]. It slightly improved some low-velocity results, but the full test-set performance did not show a clear advantage over the single-stage tree-based model. Therefore, it should be presented as a useful experimental variant, not as a definitively better solution.

5.3 Practical value of corrected Hydromast velocity measurements

The practical value of the proposed approach comes from the complementary roles of Hydromast and ADV. Hydromast is useful because it can provide continuous measurements at low cost and low energy consumption, while ADV is more suitable as a reference device but less practical for continuous long-term deployment [1], [2]. If Hydromast velocity measurements can be corrected toward ADV-like velocity values, then continuous Hydromast measurements become more useful for monitoring near-bed water velocity.

The results show that corrected Hydromast predictions were closer to ADV than raw Hydromast measurements. This is important because the raw signal contained a strong positive bias on the evaluated days. The correction model reduced this bias and produced estimates that better represented the ADV reference level.

However, the correction model should not be treated as a full replacement for ADV. ADV remains necessary as reference data for training, validation, and future recalibration. The model learns from overlapping periods where both sensors are available, so its reliability

depends on the quality and representativeness of those periods. In practical use, the model should therefore be applied only under conditions similar to the training data, or retrained and validated when new measurement conditions become available.

5.4 Reliability and validity of the results

Several aspects of the experimental design support the reliability of the results. First, model selection was based on validation data rather than test data. The test set was reserved for final reporting, which reduces the risk that reported test performance was influenced by model selection decisions.

Second, the data was split by day rather than by random rows. This is important for time-series data because neighbouring rows are highly related. A random row-level split would increase the risk of leakage, where very similar observations from the same measurement period appear in both training and testing. The day-level split makes the evaluation more realistic because the model is tested on unseen measurement days.

Third, the LSTM comparison was handled using a sequence-aligned subset. Since the LSTM requires input sequences, it naturally uses fewer rows than row-level tabular models. The sequence-aligned comparison therefore made the comparison fairer.

Fourth, the evaluation used multiple metrics: MAE, RMSE, MBE, and R^2. MAE gives an interpretable average error, RMSE gives more weight to larger errors, MBE shows systematic overestimation or underestimation, and R^2 describes explained variation [11], [12]. Visual plots also supported the numerical findings by showing that corrected predictions reduced the large offset of raw Hydromast.

Despite these strengths, the results should be interpreted carefully. The validation and test sets contained only four days each, so the results are useful but should not be treated as final proof of general performance under all possible measurement conditions. Broader validation would be needed before claiming reliability across different sites, seasons, sensors, or hydrodynamic regimes.

5.5 Limitations

The first limitation is the intermittent availability of ADV measurements. Since supervised learning requires paired Hydromast inputs and ADV target values, model training and evaluation are limited to periods where both sensors overlap. Outside ADV measurement windows, there is no direct reference target for supervised learning.

The second limitation is the small number of validation and test days. Although the row count is large because of high temporal resolution, the number of independent days is limited. Good performance on four test days does not guarantee similar performance under all other conditions.

The third limitation is the difference between training and validation/test target distributions. Chapter 4 showed that the validation and test periods contained lower ADV velocity values than the training set. This affects interpretation because models were partly evaluated under a distribution different from the training distribution. The relatively strong median baseline and underprediction in higher-velocity regions are connected to this issue.

The fourth limitation is weaker high-velocity performance. The velocity-bin analysis showed that correction models tended to underpredict higher velocities. This does not invalidate the main results, because the models still improved overall performance, but it shows that correction quality was not equal across all velocity ranges.

The fifth limitation is that only the H05 Hydromast device was used for supervised correction. Because individual sensors may differ in calibration, placement, and behaviour, the results should not be assumed to apply directly to all Hydromast devices without further testing.

Finally, the correction model is statistical. It learns patterns between Hydromast-based features and ADV velocity, but it does not fully explain the physical cause of every sensor difference. Feature importance results show predictive usefulness, not direct physical causality. Therefore, the model should be interpreted as an empirical correction tool rather than a complete physical model.

5.6 Future work

Future work should first extend the evaluation to more measurement days. A larger validation and test period would make it possible to assess whether the correction model remains reliable under more varied hydrodynamic conditions.

Second, future experiments should include more diverse velocity conditions. The current results show strong overall improvement, but high-velocity correction remains weaker. Additional training examples from higher-velocity periods could help reduce underprediction and improve robustness across the full velocity range.

Third, the approach should be tested on other Hydromast devices, not only H05. This would show whether the correction method is sensor-specific or whether it can be transferred across devices after recalibration. If device-specific differences are large, separate models or device-identification features may be needed.

Fourth, future work could add uncertainty estimation. Instead of producing only a single corrected velocity value, the model could also estimate confidence intervals or prediction uncertainty. This would help users distinguish between more and less reliable corrected predictions.

Fifth, real-time correction should be tested. Since Hydromast can provide continuous measurements, an important next step is to evaluate whether the correction model can run efficiently in a near-real-time workflow.

Finally, future work could compare model performance across different ADV-derived target definitions. A more detailed study could examine how target smoothing affects correction quality, bias, temporal responsiveness, and high-velocity behaviour.

6 Conclusion

6.1 Summary of the problem and aim

This thesis addressed the problem that raw Hydromast velocity measurements differed from ADV reference measurements. The aim was to develop and evaluate machine learning models that correct Hydromast velocity values toward ADV-like estimates.

6.2 Summary of the methodology

ADV and Hydromast data were cleaned, synchronized, resampled, aligned, and transformed into a supervised learning dataset. Several baseline, linear, tree-based, threshold-aware, and LSTM models were evaluated using MAE, RMSE, MBE, and R^2.

6.3 Main findings

The results showed that raw Hydromast strongly overestimated ADV velocity. All correction models improved over the raw baseline. The best overall results came from tabular tree-based models, while the threshold-aware model was useful but not clearly superior. The LSTM improved over raw Hydromast but did not outperform tree-based models.

6.4 Answers to research questions

The results show that machine learning models reduced the error and bias of Hydromast velocity measurements compared with the raw Hydromast baseline.

The best overall performance came from tree-based models: the threshold-aware model achieved the best validation MAE, while the single-stage tree-based model achieved the best test MAE, with only a very small difference between them.

In low-velocity conditions, both tree-based and threshold-aware models reduced error, although the threshold-aware model showed only a small advantage and was not clearly better on the full test set.

6.5 Final conclusion and future work

The thesis aim was achieved within the defined scope. The practical conclusion is that Hydromast can remain the continuous measurement device, while ADV can provide reference measurements during selected periods. Machine learning can then use these reference periods to improve Hydromast velocity measurements. Future work should test the approach on more measurement days, more diverse flow conditions, other Hydromast devices, high-velocity cases, uncertainty estimation, and real-time correction.

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